

GlucoTwin — A Digital Twin for Proactive Glucose-Spike Risk Management — Architecture

Historical / EHR → Dynamic stream → Feature fusion → Digital Twin state → AI/ML → 2-hour risk → Clinician dashboard → Human review

Validation and monitoring: patient-level split; ROC-AUC, PR-AUC, precision, recall and F1; drift and calibration checks.

Safety boundary: synthetic data • risk stratification only • no diagnosis/prescribing • clinician review required